

Vectors | HW

1/16/26

1 a) $\begin{pmatrix} 9 \\ 7 \end{pmatrix} = \begin{pmatrix} 2 \\ 8 \end{pmatrix} + \begin{pmatrix} 3 \\ -1 \end{pmatrix}$

6) $12 + k = \begin{pmatrix} 1+3k \\ 4-k \end{pmatrix}$

$1+3k = a$
 $4-k = a$
 $-3+4k = 0$
 $k = \frac{3}{4}$

c) $ap + bq = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

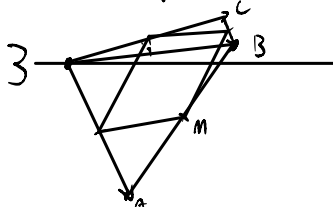
$a + 3b = 1$
 $4a - b = 0 \rightarrow a = \frac{1}{13}$
 $13b = 4$
 $b = \frac{4}{13}$

2 a) $\vec{AB} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}$
 $\vec{AC} = \begin{pmatrix} 7 \\ 2 \end{pmatrix}$
 $\vec{BC} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

6) $\vec{AB} + \vec{BC} = \vec{AC}$

c) $\frac{1}{10} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

d) $\vec{AD} = \begin{pmatrix} \frac{1}{10} - \frac{1}{10} \\ \frac{3}{10} - \frac{1}{10} \end{pmatrix}$



a) $AB = AO + OB$
 $= b - a$ (rel. a)

So midpoint = $\frac{1}{2}(b-a)$ (rel. a)

$= \frac{1}{2}(b-a) + a$ (rel. O)
 $= \frac{1}{2}(a+b)$

b) Then midpoint of BC = $\frac{1}{2}(b+c)$

midpoint of OA and OC = $\frac{1}{2}a$ and $\frac{1}{2}c$ respectively

Let midpoints of OA, AB, BC, and CO = W, X, Y, Z = $\frac{1}{2}a, \frac{1}{2}(a+b), \frac{1}{2}(b+c)$ and $\frac{1}{2}c$ respectively

Then $WX = \frac{1}{2}(a+b) - \frac{1}{2}a = \frac{1}{2}b$

$XY = \frac{1}{2}(b+c) - \frac{1}{2}(a+b) = \frac{1}{2}(c-a)$

$YZ = \frac{1}{2}(b+c) - \frac{1}{2}c = \frac{1}{2}b$

$ZW = \frac{1}{2}c - \frac{1}{2}a = \frac{1}{2}(c-a)$

4) $BC = \begin{pmatrix} 10 \\ -2 \end{pmatrix}$

$\frac{1}{2}BC = BM = \begin{pmatrix} 5 \\ -1 \end{pmatrix}$

$\therefore OM = OB + BM = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$

$|OA + AM| = | \begin{pmatrix} 9 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ -1 \end{pmatrix} | = \sqrt{24}$

$\sqrt{a^2+1} + \sqrt{(2-a)^2+1} = \sqrt{24}$

$\sqrt{a^2+1} + \sqrt{a^2-4a+5} = \sqrt{24}$

$a^2+1 + 2\sqrt{(a^2-4a+5)(a^2+1)} + a^2-4a+5 = 24$

$2a^2 - 4a - 8 + 2\sqrt{a^4 - 4a^3 + 21a^2 - 4a + 5} = 0$

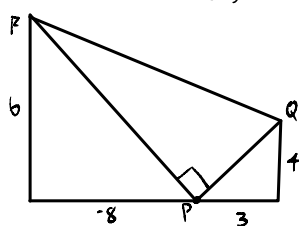
$a^2 - 2a - 4 = \sqrt{a^4 - 4a^3 + 21a^2 - 4a + 5}$

$a^4 - 4a^3 - 4a^2 + 8a + 16 = a^4 - 4a^3 + 21a^2 - 4a + 5$

$25a^2 - 20a + 11 = 0$

$(5a-2)^2 = 0$
 $a = \frac{2}{5}$

5) $\vec{PQ} = \begin{pmatrix} 3 \\ 9 \end{pmatrix}$ $\vec{QR} = \begin{pmatrix} -8 \\ 6 \end{pmatrix} + \begin{pmatrix} -3 \\ -4 \end{pmatrix}$
 $\vec{PR} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$



$|PQ| = 9$

$|PR| = \sqrt{64+36} = 10$

$|QR| = \sqrt{121+16} = 13$

If PQR is right angled, $PQ^2 + PR^2 = QR^2$

$81 + 100 = 181$

$(13)^2 = 169$

So PQR is right angled

Then RQ is the diameter of circle

so area = $\frac{1}{2} \pi r^2 = \frac{125}{4}$

Exam Questions

1 a) $d = 0.6$

b) $\begin{pmatrix} 8 \\ 6 \end{pmatrix} = k \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ $k = \frac{0.6}{3} = \frac{1}{5}$
 $\therefore a = \frac{8}{5}$

c) $\sqrt{a^2 + 0.6^2} = 1$
 $a^2 = 1 - 0.6^2$
 $a = \frac{4}{5}$

2 $A = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ gradient of A is -2 $\therefore$ gradient of B is $\frac{1}{2}$

a) $|OA| = \sqrt{1^2 + (-2)^2} = \sqrt{5}$

i) $|OB| = \sqrt{5}$

ii) $\therefore B = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ or $B = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

so $\theta = \arctan(\frac{1}{2})$ or $180 - \arctan(\frac{1}{2})$
 $= 27^\circ$ or 207° respectively.

b) $|AC| = 2$

$|OC| = OA + AC$

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$= \sqrt{5} + 2$ max

or $\sqrt{5} - 2$ min